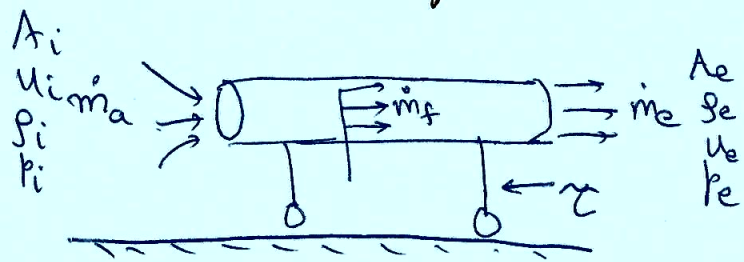
2/14/19 Ch. 3. Hypersonic Vehicle Performance

Consider airbreathing engine in 1D frictionless flow



1D mom. eqn in integral form

$$\tau = \underbrace{\dot{m}_e u_e - \dot{m}_a u_i}_{\text{Momentum thrust}} + \underbrace{p_e A_e - p_i A_i}_{\text{pressure thrust}}$$

To maximize performance, we want the nozzle to be perfectly expanded, $p_e = p_i$

From cons. of mass

$$\dot{m}_a + \dot{m}_f = \dot{m}_e$$

For perfectly expanded

$$\tau = (\dot{m}_a + \dot{m}_f) u_e - \dot{m}_a u_i$$

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Specific thrust

$$\frac{\tau}{\dot{m}_a} = \left(1 + \frac{\dot{m}_f}{\dot{m}_a}\right) u_e - u_i$$

Define fuel-air ratio $f = \frac{\dot{m}_f}{\dot{m}_a}$

$$\boxed{\frac{\tau}{\dot{m}_a} = (1+f)\, u_e - u_i}$$

For most fuels, $f \ll 1$ (0.06 for typical H_2-air)

Thrust specific fuel consumption

$$TSFC \equiv \frac{\dot{m}_f}{\tau}$$ (want to be as low as possible)

Impulse

$$I_m \equiv \int \tau \, dt$$ (recall impulse funct $I = pA(1+\gamma M^2)$)

Specific Impulse

$$I_{sp} \equiv \frac{\int \tau \, dt}{\int \dot{m}_f g \, dt}$$ ← weight flux of fuel

For const τ & $\dot{m}_f$: $I_{sp} = \frac{\tau}{\dot{m}_f g}$ → want this to be as high as possible

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Rockets $I_{sp} \sim 300$ s.

airbreathing $I_{sp} \sim 3000 - 5000$ s

Hydrocarbon combustion

$$C_xH_y + \left(\frac{2x + y/2}{2}\right)(O_2 + 3.76N_2) \longrightarrow xCO_2 + \frac{y}{2}H_2O + 3.76\left(x+\frac{y}{4}\right)N_2$$

So for one mole of C_xH_y need $\left(x+\frac{y}{4}\right)$ moles of O_2 or $4.76\left(x+\frac{y}{4}\right)$ moles of air

$$f_{st} = \left.\frac{\text{fuel mass}}{\text{air mass}}\right)_{stoic} =$$

mass = # of moles × MWT.

$M_{w,fuel} = x \times 12 + y$

$M_{w,air} = \frac{1}{4.76}(2\times16 + 3.76\times2\times14)$

$= 28.8$.

So $$\boxed{f_{st} = \frac{36x + 3y}{103(4x+y)} \text{ for } C_xH_y}$$

Define ~~equivale~~ equivalence ratio.

$$\phi = \frac{f}{f_{st}}$$

fuel lean : $\phi < 1$, have O_2 in R.H.S.

fuel rich : $\phi > 1$, have 'fuel' in R.H.S.

also CO & H_2 in add^n to pure fuel, cracked fuel, pyrolysis products

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Again for hydrocarbons ~~s~~ in air, $f_{st} \approx 0.06$

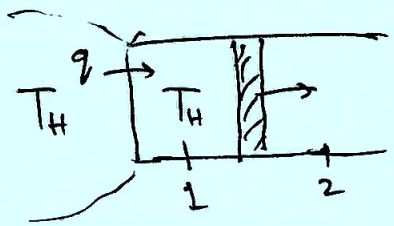
" " H_2 " " $f_{st} \approx 0.03$

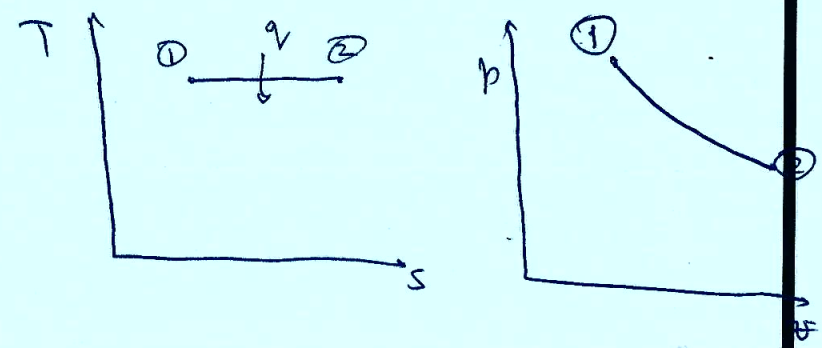
Generally, combustors are run w/ an overall $\phi < 1$ to improve I_{sp}.

Thermodynamic Cycle Review.

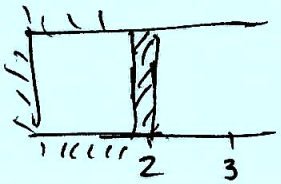
Ideal cycle : Carnot ~~cycle~~ cycle.

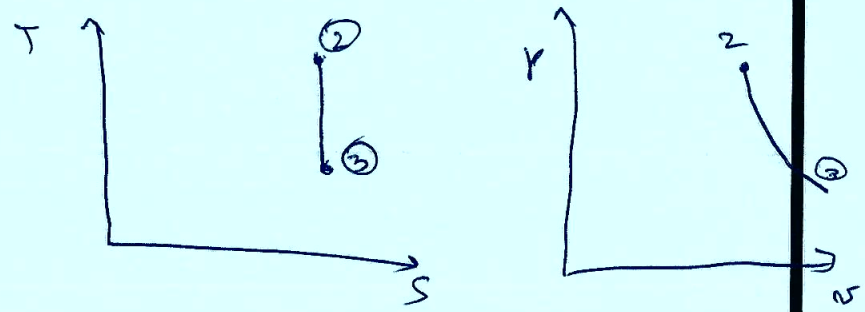
1) Isothermal heat addⁿ.





2) Isentropic expansion.





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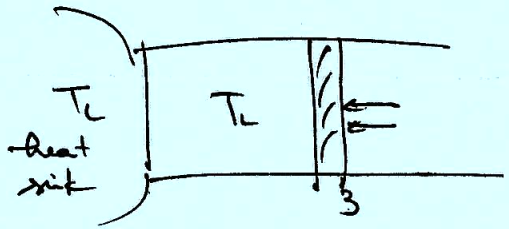
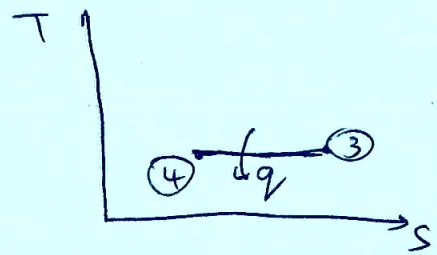
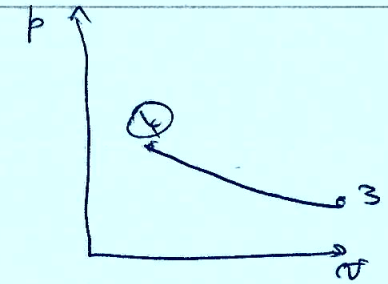
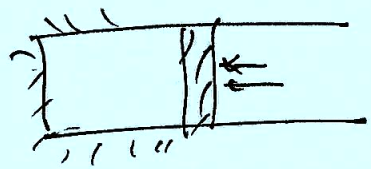
③ Isothermal heat rejection

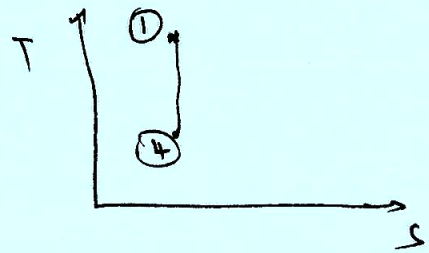


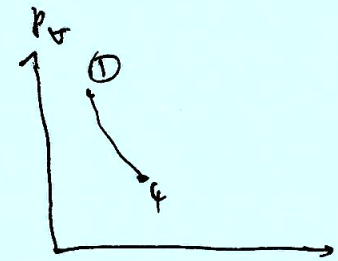




④ Isentropic comp.





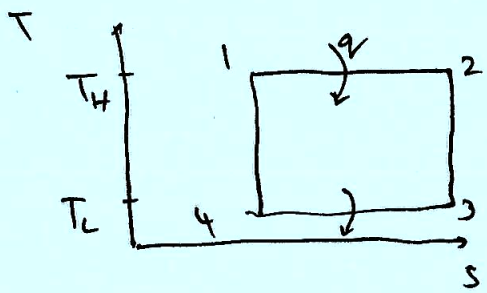


→ Reversible ⇒ No gradients ⇒ $\nabla T \sim 0$

$$\dot{q} = \nabla T \sim 0$$

$$\Rightarrow q = \dot{q}\,\Delta t$$

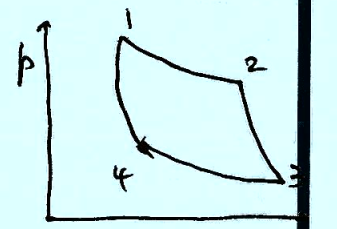
$\Rightarrow \Delta t \gg 1$ takes long time!! Not very practical.



Q : areas w/ in cycle $\left(q = \int T ds\right)$

$W = \int p\, dv$

From 1st law, $Q = W \Rightarrow \boxed{\eta_H = 1 - \frac{T_L}{T_H}}$



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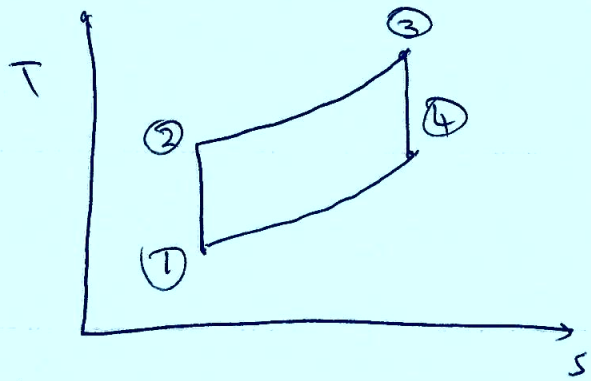
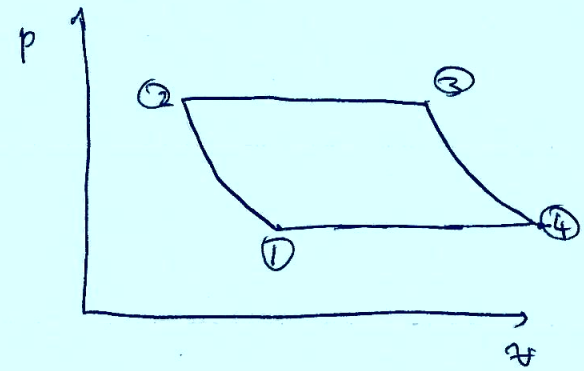
<u>Brayton cycle</u> : Replace isothermal heat addition/rejection w/ isobaric processes.

We now have 2 isobaric heat transfer processes & 2 isentropic exp/comp. process





$$\eta_{H,B} = 1 - \frac{T_1}{T_2} \qquad (T_2 < T_3 = T_H \text{ in Carnot cycle})$$

$$\Rightarrow \eta_{H,B} < \eta_{H,C} \quad (<1)$$

To increase efficiency, need higher T_2. For turbjet, want to increase compressor pr. ratio; for ram/scramjet want to increase M#.

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Efficiencies

$$\eta_{th} \equiv \frac{\text{work out}}{\text{heat in}} \equiv \frac{\text{power out}}{\text{heat flux in}}$$

Heat in engines comes from combustion of fuel. Rate of heat in $= \dot{m}_f H_{vf}$

H_{vf} = heat value $\Rightarrow$ the amount of heat liberated from combustion of a unit mass of fuel. (burned stoichiometrically).

HHV $\begin{cases} H_{vf} \approx 45 \text{ MJ/kg} & \text{typical for HCs.} \\ H_{vf}\big|_{H_2} = 120 \text{ MJ/kg} & \text{due to low mass of } H_2 \end{cases}$

HHV : H_2O in products is in liquid form.

From 1st law : rate of work out = rate of change of K.E.

$$= \dot{m}_e\left(\frac{u_e^2}{2}\right) - \dot{m}_a\left(\frac{u_0^2}{2}\right)$$

$$\rightarrow \dot{m}_e = \dot{m}_a + \dot{m}_f = (1+f)\dot{m}_a .$$

$$\Rightarrow \text{Work out} = \dot{m}_a\left((1+f)\frac{u_e^2}{2} - \frac{u_0^2}{2}\right)$$

$$\Rightarrow \eta_{th} = \frac{\dot{m}_a\left((1+f)\frac{u_e^2}{2} - \frac{u_0^2}{2}\right)}{\dot{m}_f H_{vf}} = \frac{(1+f)\frac{u_e^2}{2} - \frac{u_0^2}{2}}{f H_{vf}}$$

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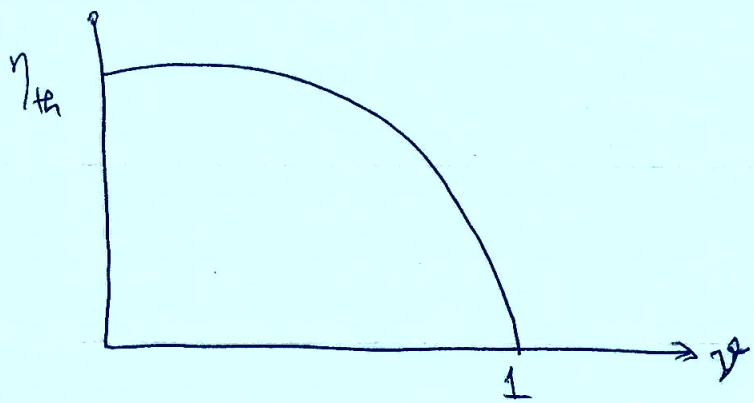
Know that to generate positive thrust, $u_e > u_0$ for all arbitrary engines.

$$\vartheta \equiv \frac{u_0}{u_e}$$

$$\eta_{th} = \frac{(1+f) - \vartheta^2}{2f\, H_{pr} v_f / u_e^2}$$



Propulsive efficiency

$$\eta_p \equiv \frac{\text{thrust power}}{\text{rate of mech work out}}$$

Thrust power $= \tau u_0$

$$\eta_p = \frac{\tau u_0}{\dot{m}_a\left((1+f)\frac{u_e^2}{2} - \frac{u_0^2}{2}\right)}$$

$$\tau = \dot{m}_a\left((1+f)u_e - u_0\right) \quad \text{perfectly expanded.}$$

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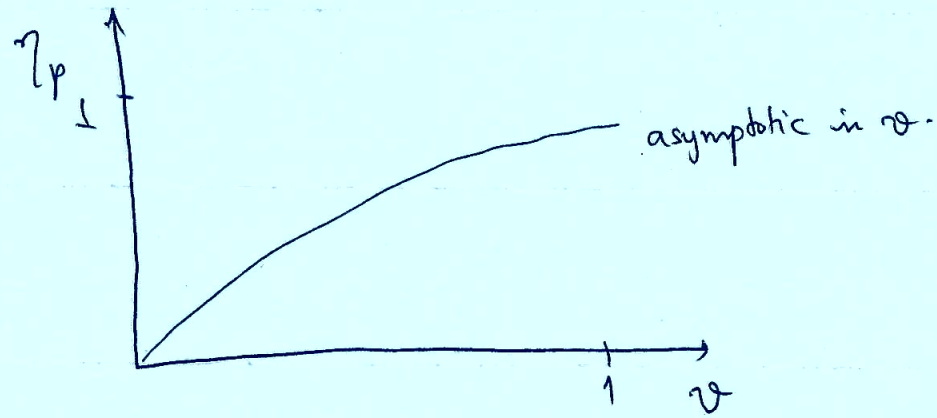
$$\eta_p = \frac{u_0\left((1+f)u_e - u_0\right)}{(1+f)\frac{u_e^2}{2} - \frac{u_0^2}{2}}$$

for $f \ll 1$ $\quad \eta_p \approx \dfrac{2}{\frac{u_e}{u_0}+1}$



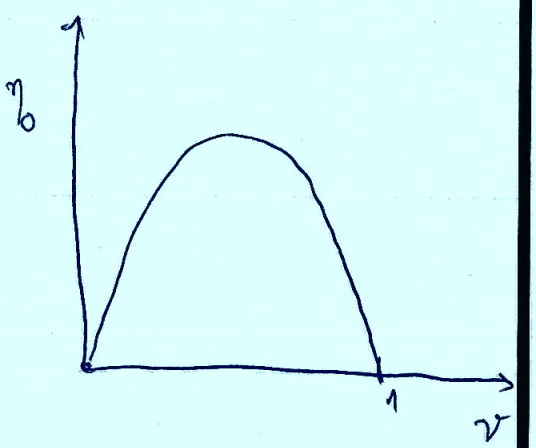
Overall efficiency.

$$\eta_o = \eta_{th} \cdot \eta_p$$

$$\eta_o = \frac{\text{Thrust power}}{\text{heat flux in}} = \frac{\tau u_0}{\dot{m}_f H_{fvf}}$$

$$\eta_o = \frac{(1+f)u_e u_0 - u_0^2}{f H_{vf}}$$

$$= \frac{(1+f)\vartheta - \vartheta^2}{f H_{vf}\, u_e^2}$$



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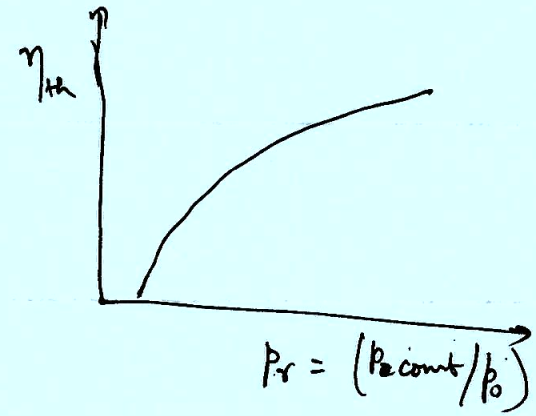
Know that $\eta_{th} \uparrow$ w/ T_2 $\left(\eta_{th} = 1 - \frac{T_1}{T_2}\right)$ & T_2 scales w/ p_2, the pr ratio

For isentropic compressor, w/ CPG

$$\frac{p_2}{p_1} = \left(\frac{T_2}{T_1}\right)^{\gamma/\gamma-1}$$

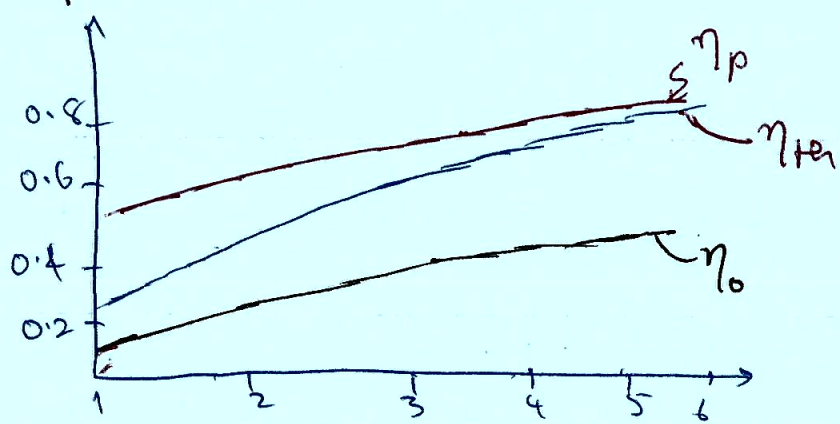
$$\eta_{th} = 1 - \frac{1}{(p_2/p_1)^{\frac{\gamma-1}{\gamma}}}$$



For non-turbomachinery $P_r = \frac{p_{comb}}{p_0} \sim M^2$

Eg: $P_r = \frac{2\gamma}{\gamma+1} M_0^2 - \frac{\gamma-1}{\gamma+1}$ for const area flow w/ N-S.

So for η_{th}, η_p, η_o as a $f(M_0)$



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